

|                                                                                                             |                                                                                                                                                                                                                                                  |                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
|  <p>दु.क्र ०२०२४४७६९३८</p> |  <p>महाराष्ट्र शासन<br/>शालेय शिक्षण व क्रीडा विभाग<br/>राज्य शैक्षणिक संशोधन व प्रशिक्षण परिषद, महाराष्ट्र<br/>७०८ सदाशिव पेठ, कुमठेकर मार्ग, पुणे ४११०३०.</p> |  <p>स्वातंत्र्याचा अमृत महोत्सव</p> |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|

### प्रश्नपेढी (Question Bank) 2024

इयत्ता:- बारावी

माध्यम:- इंग्रजी

विषय:- रसायनशास्त्र (Chemistry)

सूचना-

1. सदर प्रश्नपेढी ही १००% अभ्यासक्रमावर तयार करण्यात आली आहे.
2. सदर प्रश्नपेढीतील प्रश्न हे अधिकच्या सरावासाठी असून प्रश्नसंचातील प्रश्न बोर्डाच्या प्रश्नपत्रिकेत येतीलच असे नाही, याची नोंद घ्यावी.

## INDEX

| Sr. No. | Name of Chapter                          | Page no. |
|---------|------------------------------------------|----------|
| 1.      | Solid State                              | 3        |
| 2.      | Solutions                                | 6        |
| 3.      | Ionic Equilibria                         | 9        |
| 4.      | Chemical Thermodynamics                  | 12       |
| 5.      | Electrochemistry                         | 16       |
| 6.      | Chemical Kinetics                        | 20       |
| 7.      | Elements of Groups 16, 17 and 18         | 24       |
| 8.      | Transition and Inner transition elements | 27       |
| 9.      | Coordination compounds                   | 30       |
| 10.     | Halogen Derivatives                      | 33       |
| 11.     | Alcohols, Phenols and Ethers             | 37       |
| 12.     | Aldehydes, Ketones and Carboxylic acids  | 40       |
| 13.     | Amines                                   | 43       |
| 14.     | Biomolecules                             | 46       |
| 15.     | Introduction to Polymer Chemistry        | 49       |
| 16.     | Green Chemistry and Nanochemistry        | 52       |

## Chapter-1

### Solid State

Marks 3 with option 5

#### Multiple choice questions (1 mark)

- Which among the following solid does not show Frenkel defect.  
a. **NaCl** b.  $\text{CaF}_2$   
c.  $\text{AgI}$  d.  $\text{ZnS}$
- Which of the following crystal systems contain four Bravais lattices?  
a. cubic b. tetragonal **c. orthorhombic** d. monoclinic
- The number of spheres surrounded by octahedral void is---  
a. 3 b. 4 **c. 6** d. 8
- The amorphous solid among the following is  
a. copper sulphate b. magnesium **c. tar** d. diamond
- A paired cation-anion vacancy is called.....  
a. **Schottky defect** b. Frenkel defect c. impurity defect d. vacancy defect
- The unit cell of a simple cubic system has atoms at the eight corners. Hence, number of atoms in an unit cell is  
a. **1** b. 2 c. 3 d. 8
- In crystal lattice formed by primitive unit cell, the space occupied by particles is  
a. 32% b. 47.6% **c. 52.4%** d. 68%
- The coordination number of spheres in hcp lattice in three dimension is  
a. 2 b. 4 c. 6 **d. 12**
- A compound is made up of two elements X and Y and crystallizes in bcc structure. atoms of X are present at the corners of the cube. Atoms of Y are present at the center of the cube. The formula of the compound is  
a.  $\text{X}_2\text{Y}$  **b. XY** c.  $\text{XY}_2$  d.  $\text{X}_2\text{Y}_3$
- Sodium crystallizes in bcc structure. If the edge length of unit cell is  $4.3 \times 10^{-8} \text{ cm}$ , the radius of Na atom is  
a.  **$1.86 \times 10^{-8} \text{ cm}$**  b.  $1.52 \times 10^{-8} \text{ cm}$   
c.  $2.15 \times 10^{-8} \text{ cm}$  d.  $4.3 \times 10^{-8} \text{ cm}$

### Very short answer questions (1 Mark)

1. Write the effect on density of a substance in the Frenkel defect?
2. Write an example of a diamagnetic substance.
3. Give one property common to both hcp and ccp crystal lattices.
4. Write the relationship between the radius of the atom and edge length of the fcc unit cell.
5. Draw diagram of bcc unit cell.
6. Find the number of tetrahedral voids formed if the number of atoms in a crystal is  $N/2$ .
7. Write the percentage of empty space in the bcc lattice.
8. If the total volume of a simple cubic unit cell is  $6.817 \times 10^{-23} \text{ cm}^3$ , what is the volume occupied by particles in the unit cell? **(Ans:  $3.57 \times 10^{-23} \text{ cm}^3$ )**
9. Calculate the number of octahedral voids formed in 0.5 mol of a compound forming hcp structure **(Ans:  $3.011 \times 10^{23}$ )**
10. Write the number of atoms in the fcc unit cell.
11. Write the Coordination number of atoms in the fcc crystal lattice.
12. Write the space occupied in the bcc crystal lattice.

### Short answer questions (Type- I) (2 Marks)

1. Explain the terms: a) Isomorphism b) Polymorphism with examples
2. Classify the following solids as molecular, ionic, covalent and metallic solids. Pb,  $\text{MgF}_2$ ,  $\text{SO}_2$  and quartz
3. Explain vacancy defect with diagram.
4. Calculate the number of unit cells in 0.3 g of a species having density of  $8.5 \text{ g/cm}^3$  and unit cell edge length  $3.25 \times 10^{-8} \text{ cm}$ . **(Ans:  $1.028 \times 10^{21}$ )**
5. A compound crystallizes in bcc structure. What is the unit cell edge length if the diameter of its atom is 120 pm? **(Ans: 138.6 pm)**
6. Distinguish between crystalline solids and amorphous solids.
7. When gold crystallizes, it forms fcc unit cells. The unit cell edge length is 408 pm. Calculate the density of gold. Molar mass of gold is 197 g/mol. **(Ans:  $19.27 \text{ g/cm}^3$ )**
8. Explain the terms: a) Substitutional impurity defect b) Interstitial impurity defect.
9. Define: Schottky defect and write the conditions for the formation of Schottky defect.

10. Define a) Valence band b) forbidden zone

### Short answer questions (Type- II) (3 Marks)

1. Calculate the packing efficiency for bcc lattice.
2. In case of hcp structure, how are spheres in the first, second and third layers arranged?
3. A substance crystallizes in fcc structure. The unit cell edge length is 367.8pm. Calculate the molar mass of the substance if its density is  $21.5 \text{ g/cm}^3$ .  
(Ans:  $161.1 \text{ g/mol}$ )
4. The unit cell of Na is bcc and its density is  $0.97 \text{ g/cm}^3$ . What is the radius of a sodium atom if the molar mass of Na is  $23 \text{ g/mol}$ ?  
(Ans:  $1.86 \times 10^{-8} \text{ cm}$ )
5. Write classification of non-stoichiometric point defects. Explain with a diagram the metal deficiency defect.
6. Explain with one example each, the diamagnetic, paramagnetic and ferromagnetic substances.
7. Write the relationship between 'a' and 'r' for sc, bcc and fcc unit cells.
8. Define: a. Diamagnetic solids b. Paramagnetic solids c. Ferromagnetic solid
9. Write the formula to calculate packing efficiency and Derive relationship between molar mass, density of the substance and unit cell edge length.

### Long answer questions (4 marks)

1. What are non stoichiometric point defects? Explain with a diagram the formation of F-centers.
2. Write the classification of stoichiometric point defects. What is a substitutional impurity defect? Explain solid solutions of metals and vacancy through aliovalent cations.
3. Derive the relationship between density of substance, its molar mass and the unit cell edge length. Explain how you will calculate the number of particles, and number of unit cells in  $x \text{ g}$  of metal.
4. Define Bravais lattices. Niobium forms bcc structure. The density of niobium is  $8.55 \text{ g/cm}^3$  and the edge length of the unit cell is 330.6 pm. How many atoms and unit cells are present in 0.5 g of niobium?  
(Ans: No. of atoms-  $3.237 \times 10^{21}$  and No. of unit cell-  $1.618 \times 10^{21}$ )
5. Define: a. Conductors b. Insulators c. Semiconductors d. Doping
6. a) Write characteristics of covalent network crystals  
b) Mention types of unit cells

## Chapter-2

### Solutions

Marks 4 with option 6

#### Multiple choice questions (1 Mark)

- Which of the following example is not the state of solid solute and liquid solvent.
  - sea water
  - benzoic acid in benzene
  - sugar in water
  - Chloroform in nitrogen**
- The solubility of a gas in water
  - decreases with increase in temperature**
  - increases with increase in temperature
  - decreases with decrease in temperature
  - is not affected by temperature
- The units of Henry's law constant is—
  - $\text{bar dm}^3 \text{mol}^{-1}$
  - $\text{mol L}^{-1} \text{bar}^{-1}$**
  - $\text{L mol}^{-1} \text{bar}^{-1}$
  - $\text{bar L}^{-1} \text{mol}^{-1}$
- The colligative properties of solutions
  - depend on nature of solute particles
  - do not depend on number of solute particles
  - do not depend on dissociation of solute in solvent
  - depend on number of solute particles**
- Which of the following solutions /solvent has maximum vapor pressure ?
  - 1M copper sulphate solution
  - pure solvent water**
  - 0.5M copper sulphate solution
  - 2M copper sulphate solution
- According to Raoult's law, relative lowering of vapor pressure of solution containing dissolved non-volatile solute
  - is equal to mole fraction of solvent
  - is equal to mole fraction of solute**
  - does not depend on mole fraction of solute
  - is equal to molality of solution
- Freezing point depression constant of a solvent is
  - inversely proportional to molality of solution
  - directly proportional to molarity of solution
  - directly proportional to molality of solution
  - expressed in  $\text{K kg mol}^{-1}$**
- Which of the following statements is applicable for 0.1M urea solution and 0.1M sucrose solution?
  - osmotic pressure of urea solution is greater than that of sucrose solution
  - osmotic pressure of sucrose solution is greater than that of the urea solution
  - sucrose solution is not isotonic with urea solution
  - both the solutions have the same osmotic pressure**
- The Henry's law constant of a gas is  $6.7 \times 10^{-4} \text{ mol}/(\text{L bar})$ . Its solubility when the partial pressure of the gas at 298 K is 0.65 bar is..
  - $4.355 \times 10^{-4} \text{ mol/L}$**
  - $4.355 \times 10^{-2} \text{ mol/L}$

c.  $2.225 \times 10^{-6}$  mol/L

d.  $2.225 \times 10^{-2}$  mol/L

### Very short answer questions (1 Mark)

1. Write the SI unit Henry's law constant.
2. Write the SI unit of ebullioscopic constant.
3. Write the effect of dissolution of a nonvolatile solute on the freezing point of solvent.
4. Write the expression for relative lowering of vapor pressure.
5. Write SI unit of cryoscopic constant.
6. Write a formula to calculate molar mass of solute from boiling point elevation.
7. Write a type of solution that exhibits positive deviations from Raoult's law?
8. What is the enthalpy change and volume change on mixing of two components forming an ideal solution?
9. The vapor pressures of pure liquids A and B are 0.600 bar and 0.933 bar respectively, at a certain temperature. What is the mole fraction of solute when the total vapor pressure of their mixture is 0.8 bar?  
(Ans:0.6004)
10. The vapor pressure of a pure liquid is 0.043 bar at a certain temperature. When a nonvolatile solute is dissolved into it, the vapor pressure of the solution is found to be 0.041 bar. What is the relative lowering of vapor pressure? (Ans:0.0465)

### Short answer questions (Type- I) (2 Marks)

1. For a very dilute solution, the osmotic pressure is given by  $\pi = n_2 RT/V$  where V is the volume in L containing  $n_2$  moles of nonvolatile solute. Establish the equation for molar mass of solute.
2. Distinguish between ideal and non-ideal solutions.
3. Write two points to explain why vapor pressure of solvent is lowered by dissolving nonvolatile solute into it.
4. In what way  $K_f$  and  $K_b$  are similar and in what way they are different?
5. Calculate total moles after dissociation in 0.1M KCl solution and 0.05M aluminum sulphate solution. Which of the two solutions will have higher freezing point depression.  
(Ans: KCl=0.2mol,  $\text{Al}_2(\text{SO}_4)_3 = 0.25$  mol. Aluminium sulfate solution has higher freezing point depression)
6. When 50 g of a nonvolatile solute is dissolved in a certain quantity of solvent, the elevation of boiling point is 2.0 K. What will be the elevation of boiling point when 30 g of solute is dissolved in the same amount of the same solvent? (Ans:1.2 K)
7. The Henry's law constant of methyl bromide ( $\text{CH}_3\text{Br}$ ), is  $0.159 \text{ mol L}^{-1} \text{ bar}^{-1}$  at  $25^\circ\text{C}$ . What is the

solubility of methyl bromide in water at 25°C and at pressure of 130 mmHg? **(Ans: 0.02755M)**

8. Explain the relationship between van't Hoff factor and degree of dissociation.
9. Derive relationship between osmotic pressure and concentration of solution.

### **Short answer questions (Type- II) (3 Marks)**

1. Derive the expression for molar mass of solute in terms of boiling point elevation of solvent.
2. Explain the phenomenon of osmosis with a suitable diagram.
3. Explain with the help of vapor pressure-temperature curves for solution and solvent, why boiling point of solvent is elevated when a nonvolatile solute is dissolved into it.
4. A solution containing 3 g of solute A ( $M=60$  g/mol) in 1L solution is isotonic with a solution containing 8.55 g of solute B in 500 mL solution. What is the molar mass of B? **(Ans:342 g/mol)**
5. The vapor pressure of a pure solvent(water) at a certain temperature is 0.0227 bar. What is the vapor pressure of a solution containing 6 g of solute ( $M=60$  g/mol) in 50 g of solvent? **(Ans:0.02188 bar)**
6. Define: Reverse osmosis.Explain the relationship between van't Hoff factor and degree of dissociation.

### **Long answer questions (4 Marks)**

1. What are non-ideal solutions? Explain with reasons and diagrams the positive and negative deviations from Raoult's law shown by non-ideal solutions.
2. Explain with vapor pressure-temperature curves that the freezing point of a solvent is lowered by dissolving a nonvolatile solute into it. Give a reason for such lowering of freezing of solvent.
3. Define following terms
  - a) Reverse Osmosis b) Semi permeable membrane c) Osmotic pressure d) Isotonic solution.
4.
  - a) Derive relationship between molar mass solute from vapour pressure lowering .
  - b) Write the names of the two solutions that exhibit negative deviation from the Raoult's law.



## Ionic Equilibria

### Chapter-03

Marks 4 - with option 06

#### Multiple choice questions (1 Mark)

i) What is the percentage dissociation of 0.1 M Solution of acetic acid?

$$[K_a(\text{CH}_3\text{COOH}) = 10^{-5}]$$

- a) 0.01%      **b) 1%**      c) 10%      d) 100%

ii) For a reaction  $\text{HCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})} \rightleftharpoons \text{H}_3\text{O}^+_{(\text{aq})} + \text{Cl}^-_{(\text{aq})}$

Which of the following is a conjugate acid-base pair?

- a)  $\text{H}_2\text{O}$  and  $\text{Cl}^-$       b)  $\text{H}_3\text{O}^{(+)}$  and  $\text{Cl}^-$   
**c)  $\text{H}_3\text{O}^{(+)}$  and  $\text{H}_2\text{O}$**       d)  $\text{HCl}$  and  $\text{H}_3\text{O}^{(+)}$

iii) In biochemical system, pH of blood in our body is maintained due to following buffer

- a)  $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$       **b)  $\text{HCO}_3^- + \text{H}_2\text{CO}_3$**   
c)  $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$       d) citric acid +  $\text{Mg}(\text{OH})_2$

iv) If 'IP' is the ionic product and ' $k_{\text{sp}}$ ' is the solubility product, precipitation of the compound will occur under the condition when.

- a)  $\text{IP} = k_{\text{sp}}$       **b)  $\text{IP} > k_{\text{sp}}$**   
c)  $\text{IP} < k_{\text{sp}}$       d)  $\text{IP} \ll k_{\text{sp}}$

v)  $\text{NH}_4\text{F}$  is a salt of weak acid HF ( $k_a = 7.2 \times 10^{-4}$ ) and weak base  $\text{NH}_4\text{OH}$

( $K_b = 1.8 \times 10^{-5}$ ), the solution of  $\text{NH}_4\text{F}$  will be

- a) slightly acidic**      b) slightly basic  
c) strongly basic      d) neutral

vi) The theory which explain amphoteric nature of water is

- a) Arrhenius theory      b) Lewis theory  
c) Ostwald theory      **d) Bronsted - Lowry theory**

vii) The  $\text{p}K_b$  of weak base BOH [ $K_b(\text{BOH}) = 1 \times 10^{-5}$ ] will be

- a) -5      **b) 5**      c) 1      d)  $10^{-5}$

viii) The pH of blood in our body is maintained at ---

a) **7.36 -7.42**

b) 7.00 -7.36

c) 7.42- 7.36

d) 7.36- 7.86

### Very short answer questions (1 Mark)

- i) Write the name of buffer which is used to maintain pH of 8 to 10 for precipitation of cations of IV group in qualitative analysis
- ii) Write the solubility product of sparingly soluble salt  $\text{Bi}_2\text{S}_3$
- iii) What is the pOH if the hydrogen ion concentration in solution is  $1 \times 10^{-3} \text{ mol dm}^{-3}$  **(Ans:11)**
- iv) Write the relationship between molar Solubility (S) and solubility product ( $k_{sp}$ ) for  $\text{PbI}_2$
- v) Write any one example of salt derived from weak acid and weak base.
- vi) Write the formula to calculate  $p^H$  of buffer solution.
- vii) Indicate the one conjugate acid-base pair in the following reaction.



viii) Calculate the  $P^{\text{OH}}$  of  $10^{-8} \text{ M}$  of  $\text{HCl}$  **(Ans:7.0414)**

- ix) Write the formula to calculate the degree of dissociation of an electrolyte.
- x) Write the name of the buffer solution used in preparation of penicillin.

### Short answer questions (Type- I) (2 Marks)

- 1) Calculate the  $p^H$  and  $p^{\text{OH}}$  of  $0.0001 \text{ M}$   $\text{HCl}$  Solution **Ans: ( $p^H = 4$  and  $p^{\text{OH}} = 10$ )**
- 2) The solubility product of  $\text{BaCl}_2$  is  $4.0 \times 10^{-8}$  what will be its molar solubility in  $\text{mol dm}^{-3}$  ? **Ans: ( $S = 1 \times 10^{-2} \text{ mol dm}^{-3}$ )**
- 3) Classify the following species into Lewis acids and Lewis bases
  - i)  $\text{Cl}^{(-)}$       ii)  $\text{NH}_4^{+}$       iii)  $\text{BCl}_3$       iv)  $\text{NH}_3$
- 4) Define the following terms
  - i) Hydrogen ion concentration      ii) Hydroxyl ion concentration
- 5) Define molar solubility. Write its unit.
- 6) Write a solubility product of the following sparingly soluble salts.
  - i)  $\text{BaSO}_4$       ii)  $\text{AgCl}$
- 7) Explain Ostwald's dilution law for weak acids.
- 8) Explain Ostwald's dilution law for weak bases.

- 9) Write any four applications of buffer solution
- 10) Define the term acid and base with an example according to Bronsted-Lowry theory.

### Short answer questions (Type- II) (3 Marks)

- 1) Define buffer solution. Explain its types.
- 2) Write one application for each of the following buffers.
  - i) citrate buffer    ii)  $\text{HCO}_3^- + \text{H}_2\text{CO}_3$     iii)  $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
- 3) Derive the equation which implies that the degree of dissociation of weak acid is inversely proportional to the square root of its concentration.
- 4) A buffer solution contains  $0.3 \text{ mol dm}^{-3} \text{ NH}_4\text{OH}$  ( $K_b = 1.8 \times 10^{-5}$ ) and  $0.4 \text{ mol dm}^{-3}$  of  $\text{NH}_4\text{Cl}$ .  
Calculate pOH of the solution. **(Ans: 4.8696)**
- 6) The solubility of AgBr in water is  $1.20 \times 10^{-5} \text{ mol dm}^{-3}$ . Calculate the solubility product of AgBr. **( Ans:  $K_{sp} = 1.44 \times 10^{-10}$  )**
- 7) Define conjugate acid-base pair. Explain autoionization of water.
- 8) Explain acidity, basicity and neutrality of aqueous solution.

### Long answer questions (4 Marks)

- 1) a) Derive the equation  $p^H + p^{OH} = 14$     b) Distinguish between strong electrolyte and weak electrolyte
- 2) If 'S' is solubility in  $\text{mol dm}^{-3}$  and  $k_{sp}$  is the solubility product. Then write the relation between them for the  $\text{CaF}_2$  and  $\text{BaSO}_4$   
Calculate the concentration of  $\text{H}_3\text{O}^+$  ion in Soft drink whose  $P^H$  is 3.5 **(Ans:  $3.162 \times 10^{-4} \text{ mol/dm}^3$ )**
- 3) Explain the amphoteric nature of water.  
Define    a) Solubility product    b) Hydrolysis of salt
- 4) Explain with example types of salts.

## Chapter-04

### Marks 6- with option 08

### Multiple choice questions (1 Mark)

- i. An intensive property amongst the following is.....
  - a) Mass
  - b) Volume
  - c) Number of moles
  - d) Temperature**
- ii. The value of  $1 \text{ dm}^3 \text{ bar}$  is .....
  - a)  $10 \text{ J}$
  - b)  $10^2 \text{ J}$**
  - c)  $10^3 \text{ J}$
  - d)  $10^{-2} \text{ J}$
- iii. The work done in the  $\text{dm}^3 \text{ bar}$  when  $200 \text{ mL}$  of ethylene gas and  $150 \text{ mL}$  of  $\text{HCl}$  gas were allowed to react at  $1 \text{ bar}$  pressure is....
  - a)  $0.10$
  - b)  $0.15$**
  - c)  $0.20$
  - d)  $0.2$
- iv. The work done in vacuum when  $300 \text{ m mole}$  of an ideal gas expands until its volume is increased by  $2.3 \text{ dm}^3$  at  $1 \text{ bar}$  pressure is.....
  - a) Zero**
  - b) One
  - c) Two
  - d) Three
- v) For an Isothermal process
  - a)  $W = -Q$**
  - b)  $\Delta U = W$
  - c)  $\Delta U = Q + W$
  - d)  $\Delta U = Q$
- vi For an Isochoric process
  - a)  $\Delta U = 0$
  - b)  $\Delta V = 0$**
  - c)  $\Delta P = 0$
  - d)  $Q = 0$
- vii. The change in internal energy in a reaction when  $2 \text{ kJ}$  of heat is released by the system and  $6 \text{ kJ}$  of

work is done on the system will be.....

- a) +3 kJ                      b) -4kJ                      c) +4 kJ                      d) - 8 kJ

### Very short answer questions (1 Mark)

- 1) Write the expression to calculate maximum work done when 1 mole of an ideal gas expands isothermally and reversibly from  $V_1$  to  $V_2$ .
- 2) Write the mathematical relation between  $\Delta H$  and  $\Delta U$  during formation of one mole of  $\text{CO}_2$  under standard conditions.
- 3) Standard enthalpy of formation of water is  $-286 \text{ kJ mol}^{-1}$ . Calculate the enthalpy change for formation of 0.018 kg of water. **(Ans: -286kJ)**
- 4) Write the expression of the first law of thermodynamics for an isothermal process.
- 5) What is the sign convention when work is done on the system by the surrounding?
- 6) Write the expression showing relation between enthalpy change and internal energy change for gaseous phase reaction
- 7) Calculate enthalpy of formation of HCl if Bond enthalpies of  $\text{H}_2$ ,  $\text{Cl}_2$  and HCl are  $434 \text{ kJ mol}^{-1}$ ,  $242 \text{ kJ mol}^{-1}$  and  $431 \text{ kJ mol}^{-1}$  respectively. **(Ans: -93kJ/mol)**
- 8) Write the condition of  $\Delta G$  for Spontaneous reaction.
- 9) Write the name of a process in which there is no exchange of heat between the system and its surroundings.

### Short answer questions (Type- I) (2 Marks)

- 1) Define the terms: (i) Standard enthalpy of combustion (ii) Enthalpy of sublimation.
- 2) State and explain Hess's law of constant heat summation.
- 3) Write the features of reversible processes.
- 4) Derive an expression for pressure- volume work.
- 5) The enthalpy change of the following reaction



The bond enthalpies are

|                                 |     |       |      |
|---------------------------------|-----|-------|------|
| Bond                            | C-H | Cl-Cl | H-Cl |
| $\Delta H^0/\text{kJ mol}^{-1}$ | 414 | 243   | 431  |

(Ans: 330 kJ mol<sup>-1</sup>)

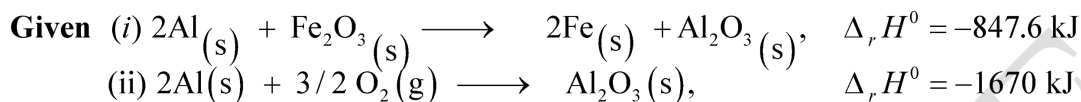
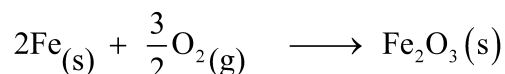
- 6) Calculate the standard enthalpy of combustion of CH<sub>4</sub>(g) if  $\Delta_f H^0(\text{CH}_4) = -74.8 \text{ kJ mol}^{-1}$ ,  
 $\Delta_f H^0(\text{CO}_2) = -393.5 \text{ kJ mol}^{-1}$  and  $\Delta_f H^0(\text{H}_2\text{O}) = -285.8 \text{ kJ mol}^{-1}$  (Ans: -890.3 kJ mol<sup>-1</sup>)
- 7) Define: a) Extensive property b) Intensive property with examples.
- 8) Write relationship between  $\Delta G$  and  $\Delta S_{\text{total}}$
- 9) Derive the relationship between  $\Delta H$  and  $\Delta U$  for a gaseous reaction.

### Short answer questions (Type- II) (3 Marks)

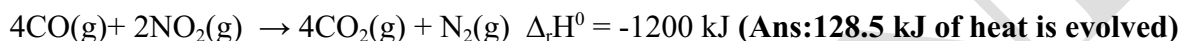
- 1) a. Define an isolated system.  
 b. Three moles of an ideal gas are expanded isothermally from 15 dm<sup>3</sup> to 20 dm<sup>3</sup> at constant external pressure of 1.2 bar, calculate the amount of work in Joules. (Ans: -600J)
- 2) Define enthalpy of fusion. Derive an expression for the maximum work.
- 3) Derive the expression  $\Delta H = \Delta U + P\Delta V$ . Write the relationship between  $Q$  and  $\Delta U$  for an isochoric process.
- 4) Define standard enthalpy of formation. Derive the relationship between standard enthalpy of reaction  
 $aA + bB \rightarrow cC + dD$  and enthalpies of formation of reactants and products.
- 5) 0.022 kg of CO<sub>2</sub> is compressed isothermally and reversibly at 298 K from initial pressure of 100 kPa when the work obtained is 1200 J, calculate the final pressure. (Ans=263.4kPa)
- 6) Define the following terms:-  
 (i) Enthalpy of vaporization (ii) Standard enthalpy of combustion.  
 Why is work done in vacuum zero
- 7) Write guidelines for writing thermochemical equations with example.

### Long answer questions (4 Marks)

- 1) Define the following terms:-  
 (i) Enthalpy of atomization (ii) Extensive properties  
 Write mathematical statement of first law of thermodynamics for following processes  
 a) Isothermal process b) adiabatic process.
- 2) Define the following terms: i) Bond Enthalpy ii) Enthalpy of ionization.  
 Calculate the standard enthalpy of the reaction. (Ans = -822.4kJ)

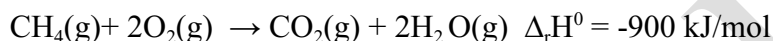


3) The amount of heat evolved when 12 g of CO reacts with NO<sub>2</sub>? The reaction is



Write an application of Hess's law.

Does the following reaction represent a thermochemical equation?

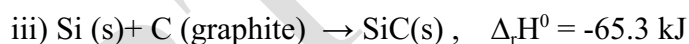
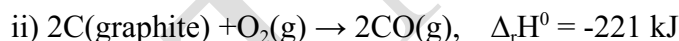


4) Classify the following into intensive and extensive properties. Pressure, volume, mass, temperature.

Define state function and write two examples of it.

5) Write sign conventions of W and Q

Calculate the standard enthalpy of the reaction (**Ans: 624.7 kJ**)



6) Define entropy and write conditions of spontaneity.

## Chapter-05

### Electrochemistry

Marks-5 with option 7

#### Multiple choice questions (1 Mark)

- i) Kohlrausch law is applicable for -----
- a) **the solution at infinite dilution**                      b) a concentrated solution  
c) concentrated as well as dilute solution                      d) aqueous solution
- ii) During electrolysis of molten NaCl, which of the following statement is Incorrect?
- a) a pale green  $\text{Cl}_2$  gas is released at anode  
b) molten silvery white sodium is deposited at cathode  
c) decomposition of NaCl into Na metal and  $\text{Cl}_2$  gas  
d) **a pale green  $\text{Cl}_2$  gas is released at anode**
- iii) SI unit of conductivity is \_\_\_\_\_
- a)  $\Omega^{-1}\text{m}^{-1}$                       b)  $\Omega\text{cm}^{-1}$                       c)  $\Omega\text{m}^{-1}$                       d)  $\Omega^{-1}\text{m}^2\text{mol}^{-1}$
- iv) In case of weak electrolyte the graph  $\Lambda$  vs  $\sqrt{c}$  is \_\_\_\_\_
- a) Linear                      b) **Non linear**                      c) Straight line passing from origin                      d) Curved
- v) In construction of Standard Hydrogen Electrode, platinum acts as \_\_\_\_\_
- a) **inert electrode**                      b) positive ion producing electrode  
c) negative ion producing electrode                      d) donor of electrons
- vi) For hydrogen gas electrode  $E_{\text{H}_2}$  is calculated through Nernst equation, where  $E^0_{\text{H}_2}$  is always
- a) **0V**                      b) 0.0592V                      c) -1.1V                      d) 1.1V
- vii) When molten ionic compound is electrolyzed a metal is formed at \_\_\_\_
- a) **Negative electrode**                      b) Positive electrode                      c) Salt bridge                      d) Electrolyte
- viii) The molar conductivity and conductivity of  $\text{AgNO}_3$  solution are  $121.4 \Omega^{-1}\text{cm}^2\text{mol}^{-1}$  and  $2.428 \times 10^{-3} \Omega^{-1}\text{cm}^{-1}$  respectively at  $25^\circ\text{C}$ . What is molar concentration of  $\text{AgNO}_3$  solution:



a) 0.02M

b) 0.2M

c) 2.0M

d) 2.2M

ix) A cell constituted by two electrodes A ( $E^0_{A+/A} = 0.35V$ ) and B ( $E^0_{B+/B} = +0.42V$ ) has value of  $E^0_{cell}$  equal to

a) 0.07V

b) -0.07 V

c) -0.77V

d) 0.77 V

x) Calculate  $E^0_{cell}$  for galvanic cell with electrodes  $Co/Co^{3+}$ ,  $Mn^{2+}/Mn$ ,  $E^0_{Mn} = -1.18V$ ,  $E^0_{Co} = 1.82 V$ .

a) -3.0V

b) + 3.0V

c) 1.36V

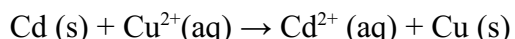
d) 0.268V

### Very short answer questions (1 Mark)

- i) Write SI unit of conductivity.
- ii) What is cell voltage?
- iii) Write a mathematical expression for Standard Cell Potential.
- iv) Write the formula to calculate molar conductivity of the given solution.
- v) Write the chemical composition present in the salt bridge.
- vi) Write the potential produced through the NICAD storage cell.
- vii) Write an equation that shows the relationship between molar conductivity and degree of dissociation of weak electrolyte.
- viii) Write the relationship between electrical conductance and molar conductivity.
- ix) Write Nernst equation.

### Short answer questions (Type- I) (2 Marks)

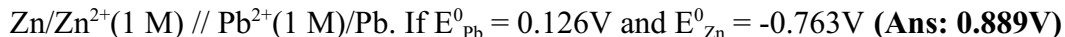
- 1) Draw a neat and labeled diagram of Standard Hydrogen Electrode.
- 2) What are the functions of a salt bridge in a galvanic cell?
- 3) Derive relation between equilibrium constant of reaction(K) and standard cell potential( $E^0_{cell}$ )
- 4) Write applications of Kohlrausch's Law.
- 5) What is the cell constant? Write its SI unit.
- 6) Mention difficulties in setting Standard Hydrogen Electrode.
- 7) What is the mass of copper metal produced at cathode during the passage of 2.03A current through the  $CuSO_4$  solution for 1 hour. Molar mass of Cu =  $63.5 g mol^{-1}$  (Ans:2.405 g)
- 8) Mercury battery provides more constant voltage than any other dry cell; Explain.
- 9) Represent the galvanic cell from following overall cell reaction



Define: Anode

10) How many moles of electrons are required for reduction of 2 moles of  $\text{Zn}^{2+}$  to Zn ? (**Ans: 4 moles**)

11) Calculate standard cell potential of following galvanic cell:



12) Draw a neat and labeled diagram of Lead accumulator.

13) Draw a neat and labeled diagram of Leclanche cell(Dry Cell)

14) What is a salt bridge? write functions of salt bridge.

15) Draw neat and labeled diagram of  $\text{H}_2\text{-O}_2$  fuel cell.

### Short answer questions (Type- II) (3 Marks)

1) State Kohlrausch law of independent migration of ions. Derive the relationship between Gibbs energy of cell reaction and cell potential.

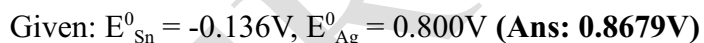
2) Write the main difference between electrolytic conductivity and molar conductivity with respect to concentration. Also write one application of electrochemical series.

3) Write three important steps required to determine molar conductivity.

4) Draw a neat and well labeled diagram of Standard Hydrogen Electrode. Write one application.

5) Define reference electrode. Write two applications of electrochemical series.

6) Calculate the voltage of the cell  $\text{Sn(s)} / \text{Sn}^{2+}(0.02\text{ M}) // \text{Ag}^+(0.01\text{ M}) / \text{Ag(s)}$  at  $25^\circ\text{C}$ .



7) Draw a well labeled diagram of a conductivity cell. Write net cell reaction involved in electrolysis of aqueous NaCl.

8) Write a mathematical formula for mole ratio. How long will it take to produce 2.415g of Ag metal from its salt solution by passing a current of 3A? Molar mass of Ag=  $107.9\text{ gmol}^{-1}$  (**Ans: 720 s or 12 min.**)

9) Write net cell reactions during discharging and recharging of lead accumulator. Write representation of SHE.

### Long answer questions (4 Marks)

1) Why is the Nickel Cadmium cell referred to as a secondary cell? Write working of NICAD storage cells. Write its applications.

2) Write relation between electrolytic conductivity and molar conductivity.

Calculate molar conductivity at zero concentration for  $\text{CaCl}_2$  and NaCl.

Given: molar ionic conductivities of  $\text{Ca}^{2+}$ ,  $\text{Cl}^-$ ,  $\text{Na}^+$  ions are respectively, 104, 76.4, 50.1  $\Omega^{-1}\text{cm}^2\text{mol}^{-1}$   
(Ans: 256.8  $\Omega^{-1}\text{cm}^2\text{mol}^{-1}$  and 126.5  $\Omega^{-1}\text{cm}^2\text{mol}^{-1}$ )

3) Calculate  $E^\circ_{\text{cell}}$  of the following galvanic cell:

$\text{Mg(s)} / \text{Mg}^{2+}(1 \text{ M}) // \text{Ag}^+(1 \text{ M}) / \text{Ag(s)}$  if  $E^\circ_{\text{Mg}} = -2.37\text{V}$  and  $E^\circ_{\text{Ag}} = 0.8\text{V}$ . Write cell reactions involved in the above cell. Also mention if cell reaction is spontaneous or not. (Ans: 3.17 V)

4) Explain construction, working in terms of cell reactions and the results of electrolysis of fused NaCl.

5) Explain Construction of Standard Hydrogen electrode (SHE), write its applications and difficulties in setting.

6) a) Write advantages of fuel cell.

b) Write oxidation half and reduction half reactions during electrolysis of molten NaCl.

## Chapter-6

### Chemical Kinetics

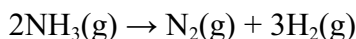
### Marks-4 with option- 6

#### Multiple choice questions (1 Mark)

- i) A First order reaction is 50% complete in 69.3 minutes. Time required for 90% completion for the same reaction is  
a) 100      b) 125 mins      c) 230 mins      **d) 230.3 mins**
- ii) Time required for 100% completion of a zero order reaction is \_\_\_\_  
**a)  $a/k$**       b)  $a/2k$       c)  $a.k$       d)  $2k/a$
- iii) Rate constant of a reaction is  $3.6 \times 10^{-3} \text{ S}^{-1}$ . The order of reaction is \_\_\_\_  
**a) First**      b) Second      c) Third      d) Zero
- iv) The rate law relates to the rate of a chemical reaction in terms of \_\_\_\_  
a) Concentration of catalyst      b) Temperature      c) Potential energy      **d) mol/L of reactants**
- v) For first order reaction the rate constant for decomposition of  $\text{N}_2\text{O}_5$  is  $6 \times 10^{-4} \text{ s}^{-1}$ . The half-life period for decomposition in seconds is \_\_\_\_  
a) 1.155      b) 11.55      c) 115.5      **d) 1155**
- vi) Order of reaction for which unit of rate constant is  $\text{mol dm}^{-3}\text{s}^{-1}$  is \_\_\_\_  
**a) 0**      b) 1      c) 2      d) 3
- vii) The rate of catalyzed reaction is larger than the uncatalyzed reaction as \_\_\_\_  
a)  $E_a (\text{catalyzed}) > E_a (\text{uncatalyzed})$       **b)  $E_a (\text{catalyzed}) < E_a (\text{uncatalyzed})$**   
c)  $E_a (\text{catalyzed}) = E_a (\text{uncatalyzed})$       d)  $E_a (\text{catalyzed}) \gg E_a (\text{uncatalyzed})$
- viii) Which of the following is a unimolecular reaction?  
a)  $2\text{HI} \rightarrow \text{H}_2 + \text{I}_2$   
**b)  $\text{N}_2\text{O}_5 \rightarrow \text{N}_2\text{O}_4 + \frac{1}{2} \text{O}_2$**   
c)  $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$   
d)  $\text{PCl}_3 + \text{Cl}_2 \rightarrow \text{PCl}_5$
- ix) Effect of catalyst in a chemical reaction is to change the  
**a) Activation energy**      b) Equilibrium concentration      c) Final products      d) Heat of a reaction

#### Very short answer questions (1 Mark)

- i) Write the unit of rate constant for the first order reaction.
- ii) Write order of the following reaction:



iii) Write molecularity of following reaction:



iv) Rate constant for the reaction  $2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$  is  $4.98 \times 10^{-4} \text{ s}^{-1}$ . Find the order of reaction?

**(Ans: First order)**

v) Write a mathematical expression for integrated rate law for zero order reaction.

vi) Write the equation for half life and rate constant of the first order reaction.

vii) Give one example of zero order reaction.

viii) For the reaction  $2\text{NO}(\text{g}) + 2\text{H}_2(\text{g}) \rightarrow \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ . The rate law is  $\text{rate} = k[\text{NO}]^2[\text{H}_2]$ .

What is the overall order of reaction?

ix) Write a unit of rate constant of zero order reaction.

x) Write only the equation of integrated rate law for the first order reaction in solutions.

xi) Write Arrhenius equation.

### Short answer questions (Type- I) (2 Marks)

- 1) What is half-life of first order reaction if time required to decrease concentration of reactants from 0.8M to 0.2M is 12 hrs. **(Ans: 6 hrs)**
- 2) Distinguish between order of reaction and molecularity.
- 3) For the reaction  $2\text{NOBr} \rightarrow 2\text{NO}_2 + \text{Br}_2$ , the rate law is  $\text{rate} = k[\text{NOBr}]^2$ . If the rate of a reaction is  $6.5 \times 10^{-6} \text{ molL}^{-1}\text{s}^{-1}$ , when the concentration of NOBr is  $2 \times 10^{-3} \text{ molL}^{-1}$ . What would be the rate constant of the reaction? **(Ans:  $1.625 \text{ Lmol}^{-1}\text{s}^{-1}$ )**
- 4) Explain pseudo first order reaction with a suitable example.
- 5) Define order of reaction with suitable examples.
- 6) Explain with the help of a potential energy diagram that the catalyst increases the rate of the reaction.
- 7) Explain the integrated rate law for zero order reactions.
- 8) How will you represent the zeroth order reaction graphically?
- 9) Explain graphically the effect of temperature on the rate of reaction.

### Short answer questions (Type- II) (3 Marks)

- 1) Derive an integrated rate law expression for first order reaction:  $\text{A} \rightarrow \text{B} + \text{C}$

- 2) Define molecularity. The rate constant of the first order reaction is  $1.386\text{min}^{-1}$ . Calculate the time required for 80% reactant to decompose? (**Ans: 1.162 min. OR 69.7 s**)
- 3) A reaction occurs in the following steps:
  - a)  $\text{NO}_2(\text{g}) + \text{F}_2 \rightarrow \text{NO}_2\text{F}(\text{g}) + \text{F}(\text{g})$  (slow)
  - b)  $\text{F}(\text{g}) + \text{NO}_2(\text{g}) \rightarrow \text{NO}_2\text{F}$  (Fast)
  - i) Write the equation of overall reaction    ii) Write the rate law    iii) Identify reaction intermediate
- 4) Define the half-life of a reaction. Write units of rate constants for:
  - a) First order reaction                      b) Zero order reaction
- 5) Write an expression for the instantaneous rate of reaction:  $2\text{N}_2\text{O}_5(\text{g}) \rightarrow 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ .  
What is the order of reaction?
- 6) Why is molecularity applicable for only elementary reactions whereas order of reaction is applicable for elementary and complex reactions? Explain with suitable examples.
- 7) For a zero order reaction molecularity can never be equal to zero. Explain.
- 8) For the reaction  $2\text{A} + \text{B} \rightarrow \text{C}$ , rate of disappearance of A  $0.076\text{ mol s}^{-1}$ .
  - a) What is the rate of formation of C?
  - b) What is the rate of consumption of B?
  - c) What is the rate of the overall reaction?

**(Ans: a.  $0.038\text{ mol s}^{-1}$  b.  $0.038\text{ mol s}^{-1}$  c.  $0.038\text{ mol s}^{-1}$ )**
- 9) Show that time required for 99.9% completion of a first order reaction is three times the time required for 90% completion.

### Long answer questions (4 Marks)

- 1) In a first order reaction  $\text{A} \rightarrow \text{B}$ , 60% of a given sample of a compound decomposes in 45 mins. What is the half-life of a reaction? Also write the rate law equation for the above first order reaction.  
**(Ans:  $t_{1/2} = 34.02\text{min}$ )**
- 2) Derive an expression for the relation between half-life and rate constant for first order reaction. The half-life period for first order reaction is 1.7 hrs. How long will it take for 20% of the reactant to disappear?  
**(Ans:  $t = 0.5475\text{ hrs. or } 32.86\text{ min}$ )**
- 3) Write one example of the reaction where order and molecularity are the same. Mention any two factors that influence the rate of chemical reaction. If for the reaction  $\text{A} \rightarrow \text{products}$ , a straight line graph passing through origin is obtained between the rate of reaction against concentration of A, what would be the order

of reaction? Why

SCERTM, PUNE

## Chapter- 07

### Elements of Group 16, 17, 18

Marks 6 with option 8

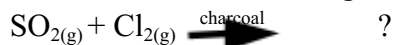
#### Multiple Choice Questions (1 Mark)

- i) In chlorous acid, the oxidation state of chlorine is.....  
(a) +2                      (b) +3                      (c) +4                      (d) +7
- ii) Acidic strength of halogen acids increases in the order of....  
(a)  $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$                       (b)  $\text{HCl} > \text{HF} > \text{HBr} > \text{HI}$   
(c)  $\text{HBr} > \text{HCl} > \text{HF} > \text{HI}$                       (d)  **$\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$**
- iii) Sulfur dioxide reacts with sodium hydroxide solution to form \_\_\_\_\_  
(a) **Sodium Sulfite**                      (b) Sodium Sulfate  
(c) Sodium hydrogen sulfite                      (d) Sodium hydrogen sulfate
- iv) The gas is evolved, when sulfuric acid reacts with copper metal \_\_\_\_\_  
(a) **Sulfur dioxide**                      (b) Sulfur trioxide  
(c) Nitrogen dioxide                      (d) Nitrogen trioxide
- v) When hot and concentrated alkali NaOH reacts with chlorine to form \_\_\_\_\_  
(a) Only Chlorate                      (b) Only Hypochlorite  
(c) **Chloride and Chlorate**                      (d) Chloride and Hypochlorite
- vi) When  $\text{SO}_2$  is passed through an aqueous solution of  $\text{I}_2$  solution becomes \_\_\_\_\_  
(a) ruby red                      (b) **colorless**                      (c) violet                      (d) yellowish green
- vii)  $\text{O}_2$  molecule is \_\_\_\_\_  
(a) ferromagnetic                      (b) diamagnetic  
(c) **paramagnetic**                      (d) ferrimagnetic
- viii) The number of covalent bonds are present in sulfuric acid?  
(a) 2                      (b) 4                      (c) 6                      (d) **8**
- ix) In Interhalogen compounds, which halogen is never the central atom?  
(a) I                      (b) **F**                      (c) Br                      (d) Cl
- x) The name of  $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$  is ---  
a) epsom salt                      (b) **gypsum**                      c) galena                      d) carnallite



### Very Short Answer Questions ( 1 Mark)

i) Complete and write the following chemical reaction.



ii) Write the name of a solution formed by passing sulfur dioxide in water.

iii) Write chemical formula of galena

iv) Why does oxygen cannot exhibit higher oxidation state?

v) The number of lone pairs of electron are present in  $\text{ClF}_5$

vi) Write the order of ionic character of halide with monovalent metal (M)

vii) Write chemical composition of cryolite.

viii) Write chemical formula of fluorapatite.

### Short Answer Questions (Type- I ) (2 Marks)

Q.1 Draw structure and name the shape of bromine trifluoride.

Q.2 Write four uses of chlorine.

Q.3 Write a balanced chemical reaction of sulfuric acid with (a) carbon (b) sulfur.

Q.4 Draw resonance hybrid structure of  $\text{SO}_2$  in two canonical forms.

Q.5 What is the action of chlorine on (a) cold and dilute sulfuric acid (b) hot and concentrated sulfuric acid.

Q.6 Elements of group 16 have lower ionization enthalpy values compared to those of group 15 elements.

Explain why?

Q.7 Write uses of dioxygen.

Q.8 a. Define: Dry bleach

b. Write the name of an element, which is a radioactive decay product of thorium and uranium.

Q.9 Write uses of Neon and Helium.

Q.10 Draw the structure and write shape of Chlorine pentafluoride.

Q.11 Write structure of ozone and sulfur dioxide.

Q.12 Explain Oxidising and bleaching properties of Chlorine.

### Short Answer Questions (Type-II) (3 Marks)

Q.1 What is oxidation state of sulfur in following

(a) Sulfurous acid      (b) Sulfuric acid      (c) Peroxy monosulfuric acid.

Q.2 Explain why fluorine shows only +1 oxidation state while other halogens show higher positive

oxidation state?

Write chemical reaction of action of  $\text{Cl}_2$  in excess  $\text{NH}_3$

Q.3 Distinguish between rhombic sulfur and monoclinic sulfur with respect to following points:

Colour, shape, melting point, density, solubility in  $\text{CS}_2$ , structure.

Q.4 Explain the trend in the following atomic properties of group 16 elements:

(a) atomic radii (b) electronegativity (c) electron gain enthalpy

Q.5 What are chalcogens ? Discuss industrial method of preparation of sulfur dioxide from zinc sulfide and iron pyrites.

Q.6 Write three physical properties and three uses of sulfuric acid.

Q.7 Explain the anomalous behavior of Oxygen with respect to i) Atomicity ii) Magnetic property  
iii) Oxidation state

Q.8 Define: Interhalogen compounds. Write general characteristics of interhalogen compounds.

Q.9 Write preparation of Potassium dichromate by using Chromite ore.

Q.10 Write order of Halogen acids with respect to a) acidic strength b) Bond dissociation enthalpy  
c) Thermal stability

### Long Answer Questions ( 4 Marks)

Q.1 Write chemical reactions in the manufacture of sulfuric acid by contact process.

Q.2 What happens when chlorine reacts with the following:

(a) Al (b) Na (c)  $\text{S}_8$  (d)  $\text{P}_4$

Q.3 Draw structure of chloric acid and chlorous acid. Discuss four points of anomalous behavior of fluorine.

Q.4 Write the structures of following oxoacids of sulfur

i) Pyrosulphuric acid ii) Peroxy mono sulphuric acid  
iii) Peroxydisulfuric acid iv) Thiosulphuric acid.

Q.5 Write the structures of following oxoacids of chlorine

i) Chloric acid ii) Hypochlorous acid iii) Chlorous acid iv) Perchloric acid.

Q.6 Draw the structures of i)  $\text{XeF}_2$  ii)  $\text{XeF}_4$  iii)  $\text{XeF}_6$  iv)  $\text{XeOF}_4$

Q.7 Write chemical equations for the following.

a) Excess of ammonia on chlorine b) Dry slaked lime on chlorine c) excess fluorine and chlorine  
d) Xenon on fluorine

## Chapter- 08

**Transition and Inner transition elements      Marks- 6 with option 8**

### Multiple Choice Questions (1 Mark)

- i) The following ion has the maximum number of unpaired electrons
- (a)  $\text{Sc}^{3+}$  (b)  $\text{Ti}^{3+}$   
(c)  $\text{Fe}^{3+}$  (d)  $\text{Co}^{2+}$
- ii) In 3d series, if nuclear charge increases, the shielding effects will \_\_\_\_
- (a) **increases** (b) decreases  
(c) first increase then decrease (d) first decrease then increase
- iii) Transition elements have more tendency to form interstitial compounds because of..
- (a) **defect in their crystal lattice** (b) they have reducing property  
(c) they have low ionization enthalpy (d) they have same atomic size
- iv) The following electronic configuration of elements shows highest oxidation state
- (a)  $3d^5 4s^2$  (b)  $3d^5 4s^1$   
(c)  $3d^{10} 4s^2$  (d)  $3d^{10} 4s^1$
- v) Zinc does NOT show variable valency because \_\_\_\_
- (a) complete 4s subshell (b) **complete d subshell**  
(c) incomplete d subshell (d) incomplete s subshell
- vi) The catalyst used for decomposition of  $\text{KClO}_3$
- (a)  $\text{ZnO}$  (b)  **$\text{MnO}_2$**   
(c)  $\text{CuO}$  (d)  $\text{K}_2\text{O}$
- vii) The atomic number of transuranium elements starts from-----
- (a) 89-103 (b) 90-103  
(c) 91-103 (d) **93-103**
- viii) The following pair of elements has half-filled d-orbitals
- (a) chromium and cobalt (b) manganese and nickel  
(c) **chromium and manganese** (d) cobalt and nickel

### Very Short Answer Questions (1 Mark)

- i) Write a formula to calculate magnetic moment.
- ii) Write the general electronic configuration of 3d series.
- iii) Write the name of the radioactive element of Lanthanoid.
- iv) Write the formula to calculate magnetic moment.
- v) Write chemical formula of ore of zinc.
- vi) Write the name of alloy formed from copper and tin.
- vii) Which alloy is used in the Fischer-Tropsch process in the synthesis of gasoline ?
- viii) Write the name of catalyst used in the hydrogenation of ethene to ethane .
- ix) Write the general electronic configuration of lanthanoids.
- x) Write the general electronic configuration of 4d series.

### Short Answer Questions (Type- I ) (2 Marks)

- Q.1) Salt of  $\text{Sc}^{3+}$  and  $\text{Ti}^{4+}$  are colorless. Explain why?
- Q.2) Write observed electronic configuration of Europium ( $Z=63$ ) and Gadolinium ( $Z=64$ ).
- Q.3) Distinguish between lanthanides and actinides.
- Q.4) Manganese in the +2 oxidation state is more stable than +3 oxidation state whereas iron is stable at +3 oxidation state than +2 oxidation state. Explain why?
- Q.5) Explain terms cast iron and wrought iron with their uses?
- Q.6) What are the causes of lanthanide contraction?
- Q.7) Calculate the spin only magnetic moment of divalent cation of a transition metal with atomic number 26.
- Q.8) Write similarities between Lanthanoids and actinoids.
- Q.9) Define: a) Ore b) Mineral
- Q.10) Define: Gangue. Write chemical composition of Haematite.
- Q.11) Write the conditions of color of transition metal ion.

### Short Answer Questions (Type-II) (3 Marks)

- Q.1) Give similarities and differences in the elements of 3d, 4d and 5d series.
- Q.2) Discuss the position of d-block elements, lanthanoids and actinoids in the periodic table.
- Q.3) Calculate magnetic moment of thorium ( $Z=90$ ). Is this element diamagnetic or paramagnetic?
- Q.4) What are interstitial compounds? write any four properties of it.

- Q.5) What are ferrous and non-ferrous alloys? Write any two uses of alloy.
- Q.6) What are rare earth elements? Write any two properties and uses of actinides.
- Q.7) Define: a) Pyrometallurgy b) Hydrometallurgy c) Electrometallurgy
- Q.8) Write chemical equations for the preparation of potassium dichromate.

**Long Answer Questions ( 4 Marks)**

- Q.1) Ground state electronic configurations of gadolinium and lawrencium are different than expected. Explain why?
- Q.2) Explain the trends in: (a) Atomic radii (b) Oxidation state of 3d elements.  
Which factors relate to the color of transition metal?
- Q.3) Define transuranium elements and d-block elements. Write two applications of lanthanides and actinides.
- Q.4) Write the chemical composition of the following minerals.  
i) Haematite ii) Chalcopyrite iii) Calamine iv) Chalcocite

## Chapter-09

### Coordination Compounds      Marks- 5 with option 7 Multiple Choice Questions (1 Mark)

- i) Amongst the following, the ambidentate ligand is.....  
a) Ethylene diamine      b) Oxalate ion  
c) Chloride ion      **d) Cyanide ion**
- ii) The charge on metal ion in  $[Fe(CN)_6]^{4-}$  is .....  
a) 2+      b) 3+  
c) 4+      d) 5+
- iii) The effective atomic number of cobalt in  $[Co(NH_3)_6]^{3+}$  is.....  
a) 33      b) 34  
c) 35      **d) 36**
- iv) The IUPAC name of  $Na_3[AlF_6]$  is.....  
a) Hexafluorosodiumaluminate      **b) Sodium hexafluoroaluminate(III)**  
c) Sodium hexafluoroaluminate(II)      d) Sodium hexafluoroaluminium (III)
- v) Hybridization of cobalt in  $[Co(NH_3)_6]^{3+}$  the complex ion is.....  
a)  $sp^3d^2$       b)  $sp^2d^3$   
**c)  $d^2 sp^3$**       d)  $d^3 sp^2$
- vi) The geometry of  $[CoF_6]^{3-}$  the complex ion is.....  
a) Trigonal bipyramidal      b) Tetrahedral  
**c) Octahedral**      d) square planar
- vii) The pair  $[Co(NH_3)_5(SO_4)]Br$  and  $[Co(NH_3)_5Br]SO_4$  exhibits..... isomerism  
a) Coordination      **b) Ionization**  
c) Linkage      d) Optical
- viii) The number of ions are produced on dissolution of the complex  $[Co(NH_3)_6]Cl_3$  in water.  
a) 6      **b) 4**      c) 3      d) 2

### Very Short Answer Questions ( 1 Mark)

1. Draw structure of Ethylenediaminetetraacetate ion.
2. Write coordination number of Fe in  $[Fe(CN)_6]^{3-}$
3. Draw the structure of oxalate ion.
4. Write oxidation number of iron in  $[Fe(CO)_5]$
5. Calculate effective atomic number of iron in  $[Fe(CN)_6]^{4-}$  complex ion
6. Write the type of isomerism exhibited by  $[Co(NH_3)_5(NO)]^{2+}$  and  $[Co(NH_3)_5ONO]^{2+}$  pair of complex ion.
7. Write the IUPAC name of  $[Fe(CN)_6]^{4-}$  ion.
8. Write the name of a platinum complex used in the treatment of cancer.
9. Write the name of hexadentate ligand which is used for treatment of lead poisoning.
10. Identify and write the counter ion present in coordination complex  $K_4[Fe(CN)_6]$  dissociates in water

### Short Answer Questions (Type- I ) (2 Marks)

- 1) Explain homoleptic and heteroleptic complexes.
- 2) Write four postulates of Werner's theory
- 3) Write one example each of bidentate and ambidentate ligands.
- 4) Distinguish between double salt and coordination complex.
- 5) Define the following terms: (i) Coordination isomer (ii) Hydrated isomers
- 6) Write two applications of coordination compound.

### Short Answer Questions (Type-II) (3 Marks)

- 1) Write classification of complexes on the basis of type of ligands with one example of each type.
- 2) Define following terms (i) Coordination isomer (ii) Hydrated isomers.

Draw structure of cis isomer of  $[Co(NH_3)_4 Cl_2]^+$

- 3) Write a formula to calculate EAN with significance of terms involved in it. Calculate EAN of  $[Fe(CN)_6]^{3-}$ .

4. Write the IUPAC name of  $[Ni(CN)_4]^{2-}$ .

Draw the geometrical isomers of following complexes  $[Pt(NH_3)_2(H_2O)_2Cl_2]$  and  $[Co(NH_3)_4 Cl_2]^+$

5. Define ligand. Explain the magnetic properties of  $[Ni(CN)_4]^{2-}$ .

6. Define (i) Anionic complex (ii) coordination number. Draw optical isomers

Of  $[Co(en)_3]^{3+}$

7. Define: Coordination number. write two examples of coordination compounds which do not obey EAN rule.

### Long Answer Questions (4 Marks)

- 1) Write oxidation state and coordination number of Co in  $[Co(NH_3)_4 Cl_2]^+$  ion. Calculate EAN of iron in  $[Fe(CN)_6]^{4-}$ . Write the IUPAC name of  $[Zn(NH_3)_4]^{2+}$ .

- 2) Explain, why  $[Co(NH_3)_6]^{3+}$  ion is low spin? Calculate number of unpaired electrons and write the geometry of  $[Co(NH_3)_6]^{3+}$ .

- 3) Answer the following with respect to  $[CoF_6]^{3-}$  ion

i) Type of hybridization

ii) Number of unpaired electrons

iii) Geometry of complex ion

iv) Magnetic property.

- 4) Explain Cis-trans isomerism in a) Square planar complexes b) Octahedral complex

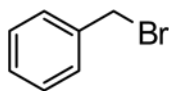


## Halogen Derivatives

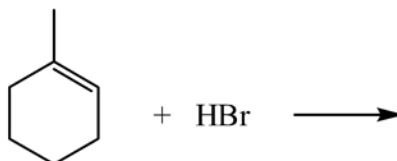
### Multiple choice questions (1 Mark)

- ### Very Short Answer Questions ( 1 Mark)

i) Write IUPAC name of the following.

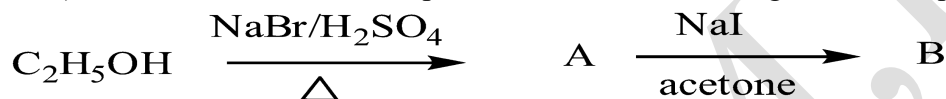


ii) Write the major product of the following reaction.



iii) Write the correct decreasing order of boiling point for  
bromomethane, chloroform, dibromomethane and bromoform .

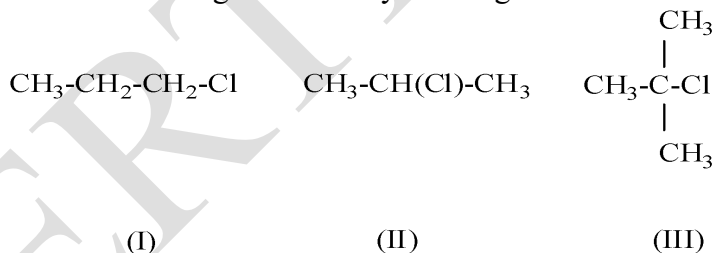
iv) Write IUPAC name of the product 'B' in the following reaction sequence.



v) Nucleophilic substitution reaction of 2,4-dinitrochlorobenzene is faster than p- nitro chlorobenzene.  
Give a reason.

vi) Write the name of the reagent used to convert alkyl halide to ester.

vii) Write the correct order of increasing ease of dehydrohalogenation.



viii) Write common name of compound having IUPAC name 1-Chloro-2,2 dimethylpropane

### Short Answer Questions (Type- I ) (2 Marks)

i) Define Optical activity. Draw the structure of DDT.

ii) Explain reactions of haloarenes with sodium metal.

iii) What is the action of chlorine on chlorobenzene? Write two uses of chloroform .

iv) Explain optical activity of 2-chlorobutane.

v) Distinguish between  $\text{S}_{\text{N}}1$  and  $\text{S}_{\text{N}}2$  mechanisms .

vi) Explain primary benzylic halide shows higher reactivity by  $\text{S}_{\text{N}}1$  mechanism than other primary alkyl halide.

- vii) Explain Wurtz reaction by taking examples of two different alkyl halides.
- viii) Explain Kharasch-mayo effect with example.
- ix) Explain optical activity of lactic acid.
- x) What is the action of following reagents on methyl bromide
  - a) bromobenzene with 'Na' metal b) 'Mg' metal in dry ether

### Short Answer Questions (Type-II) (3 Marks)

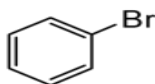
- i) Explain the factors affecting  $S_N1$  and  $S_N2$  mechanisms.
- ii) Explain alkaline hydrolysis of aqueous tert. butyl bromide.
- iii) How are the following conversions carried out?
  - a) propene to 1-iodopropane b) propene to 2-nitropropane c) benzene to biphenyl
- iv) What is Grignard reagent? How is it prepared? Why are they prepared under anhydrous condition?
- v) Write chemical equations indicating the action of following on bromobenzene.
  - a)  $\text{CH}_3\text{COCl}$  / anhy.  $\text{AlCl}_3$  b) fuming  $\text{H}_2\text{SO}_4$  c) conc.  $\text{HNO}_3$  / conc.  $\text{H}_2\text{SO}_4$
- vi) An organic compound A with molecular formula  $\text{C}_4\text{H}_{10}\text{O}$  on treatment with phosphorus pentachloride gives alkyl chloride. Alkyl chloride on treatment with Mg in presence of dry ether gives a highly reactive compound B. Compound B reacts with water to give hydrocarbon C. Alkyl chloride on treatment with Na in dry ether as a solvent gives alkane, 2,2,3,3-tetramethylbutane. Identify 'A', 'B', 'C'.
- vii) What is the action of Conc.  $\text{HNO}_3$  on chlorobenzene. Write uses and environmental effect of carbon tetrachloride

### Long Answer Questions ( 4 Marks)

- i) Write the chemical reactions for preparation of ethyl chloride from alcohols using following reagents.
  - a) Halogen acid b) Phosphorus halide c) Thionyl chloride

Which among the above method is preferred for preparation of ethyl chloride and why?
- ii) What is dehydrohalogenation? State the rule for formation of preferred products of dehydrohalogenation. Predict all the alkenes that would be formed by dehydrohalogenation of the following alkyl halide.
  - a) 2-chloro-2-methylbutane b) 3-bromo-2,2,3-trimethylpentane

iii) Observe the following compounds and answer the questions given below.



(I)



(II)

- a) Write the type of halides   b) Explain the nature of the C – Br bond in both of these halides.  
c) Which of these compounds will undergo aqueous alkaline hydrolysis readily?

Write the reaction in support of your answer.

iv) Convert the following

- a) Grignard reagent into alkane  
b) 2,4,6 -trinitrochlorobenzene into 2,4,6-trinitro phenol  
c) primary alcohol into primary alkyl iodide  
d) Ethene into ethyl bromide

## Chapter- 11

# Alcohols, Phenols and Ethers

## Marks 4 with option 6

### Multiple choice questions (1 Mark)

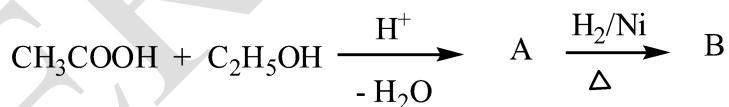
- i) In a carbinol system of nomenclature tert.butyl alcohol is named as....
- a) trimethyl carbinol                      b) dimethyl ethyl carbinol  
c) methyl carbinol                          d) ethyl carbinol
- ii) Which of the following pair of reagent is used for conversion of carboxylic acid to alcohol
- a)  $\text{LiAlH}_4 / \text{H}_3\text{O}^+$                       b)  $\text{H}_2 / \text{Ni}$  – heat  
c)  $\text{B}_2\text{H}_6 / \text{H}_2\text{O}_2, \text{OH}^-$                       d)  $\text{H}_2 / \text{Pd}$
- iii) Reaction between Grignard reagent and aldehyde other than formaldehyde leads to formation of
- a) primary alcohol                      b) secondary alcohol  
c) tertiary alcohol                      d) optically active alcohol
- iv) Complete oxidation of ethyl alcohol using  $\text{K}_2\text{Cr}_2\text{O}_7/\text{dil H}_2\text{SO}_4$  leads to formation of...
- a) acetaldehyde                      b) acetic acid  
c) ethane                      d) acetone
- v) When vapors of tert.butyl alcohol are passed over hot copper, it gives ....
- a) butanal                      b) butanoic acid  
c) butanone                      d) isobutylene
- vi) Reaction between hot conc. HI and anisole gives ....
- a) phenol and methyl iodide                      b) iodobenzene and methyl alcohol  
c) iodobenzene and methyl iodide                      d) phenol and methyl alcohol
- vii) The reagents used to convert phenol to 2,4,6-tribromophenol is ...
- a)  $\text{Br}_2 / \text{CS}_2$                       b)  $\text{Br}_2 / \text{CCl}_4$   
c)  $\text{Br}_2 / \text{H}_2\text{O}$                       d) KBr aq.
- viii) The correct IUPAC name of Hydroquinone is —
- a) quinol                      b) benzene -1,4 diol  
c) benzene-1,2 diol                      d) benzene-1,3 diol

### Very Short Answer Questions ( 1 Mark)

- i) Why do phenols give deep coloration with neutral ferric chloride?
- ii) Write the name of a valuable byproduct formed in preparation of phenol by using cumene.
- iii) Write IUPAC name of crotonyl alcohol.
- iv) Write the structure of the major product of hydroboration-oxidation of propene.
- v) Write the reaction for the preparation of aspirin from salicylic acid.
- vi) Write the name of major product when anisole reacts HI at 398 K
- vii) What is the action of atmospheric oxygen on ethers?
- viii) Draw intramolecular hydrogen bonding structure in o-nitrophenol .
- ix) Write the name of the product formed, when phenol reacts with neutral ferric chloride.

### Short Answer Questions (Type- I ) (2 Marks)

- i) What is the action of following reagents on pent-3-enal, write a complete chemical equation.
  - a)  $\text{H}_2 / \text{Ni}$
  - b)  $\text{LiAlH}_4 / \text{H}_3\text{O}^+$
- ii) Write the reactions involved in preparation of phenol from aniline.
- iii) Write Kolbe's reaction.
- iv) Write the name of reagents used to convert phenol into
  - a) picric acid
  - b) p-benzoquinone
- v) Write the structure of 'A' and 'B' in the following reaction sequence.



- vi) Write chemical equations, when acetyl chloride reacts with following reagents
  - a) ethanol
  - b) phenol
- viii) What is the action of acetic anhydride on
  - a) ethanol
  - b) salicylic acid

### Short Answer Questions (Type-II) (3 Marks)

- i) What is the action of following reagents on phenol at low temperature.
  - a) dil.  $\text{HNO}_3$
  - b) conc.  $\text{H}_2\text{SO}_4$
  - c)  $\text{Br}_2/\text{CS}_2$
- ii) Explain with reactions, the action of Lucas reagent on primary, secondary and tertiary alcohols.
- iii) How will you bring about the following conversions?

- a) isopropyl alcohol to acetone
  - b) 2-methyl propan-2-ol to 2-methylpropene
  - c) acetone to 2-methylpropan-2-ol
- iv) Compound 'A' with molecular formula  $C_6H_5Cl$  is fused with NaOH at high temperature under pressure to give compound 'B'. Compound 'B' on treatment with dil.HCl gives compound C having characteristic carbolic odor. Write the chemical equations in support of this. Name the process and give uses of compound C.
- v) Write the preparation of ethanol from methyl magnesium iodide. Write the reaction between ethanol and acetic anhydride.
- vi) Write the reactions for preparation of carbolic acid from aniline. What is the action of conc.  $H_2SO_4$  on carbolic acid at 373 K.
- vii) Write chemical equations for the preparation of phenol from benzene sulphonic acid.

### Long Answer Questions ( 4 Marks)

- i) Write chemical reactions to convert phenol into salicylaldehyde. Write the name of the reaction.  
What happens if  $CCl_4$  is used instead of  $CHCl_3$  in the above reaction.
- ii) An organic compound gives hydrogen on reaction with sodium metal. It forms an aldehyde with molecular formula  $C_2H_4O$  on oxidation with pyridinium chlorochromate. Give the chemical equations in support of these observations.  
Explain the fact that in alkyl aryl ethers, alkoxy group is ring activating and ortho/para directing towards electrophilic aromatic substitution.
- iii) How will you prepare diethyl ether by dehydration of alcohols? What are the limitations to prepare ether by this method? What is the action of following on diethyl ether?
- a) dil.  $H_2SO_4$
  - b)  $PCl_5$
- iv) Write the chemical reactions of the following.
- a) anisole reacts with HI
  - b) t-butyl chloride reacts with sodium ethoxide
  - c) t-butyl alcohol heated with 'Cu' at 573 K
  - d) benzene diazonium chloride on hydrolysis

**Aldehydes, Ketones and carboxylic acids.      Marks - 06 with option 08**

i) Which of the following is the product of reaction of ethane nitrile with Grignard reagent followed by acid hydrolysis?

a) **a ketone**

b) an aldehyde

c)an acid

d) an acid chloride

ii) ----- reaction is used to synthesize straight alkyl substituted benzenes.

a) Etard

b) Rosenmund reduction

c) Stephen reaction

**d) Wolf Kishner reduction**

iii) Ketones can **NOT** be converted into carboxylic acid using\_\_\_\_\_.

a) acidic  $\text{KMnO}_4$

b)  $\text{CrO}_3$

c) **dil. $\text{HNO}_3$**

d) Hot and conc.  $\text{HNO}_3$

iv) Correct order of acid strength for

i) acetic acid ii) fluoroacetic acid iii) 4-Nitrobenzoic acid iv) 4-Methyl benzoic acid is.....

a) i > ii > iii > iv

b) **ii > iii > iv > i,**

c) iii > iv > i > ii

d) iv > i > ii > iii

v) Addition of sodium bisulphite to ethanol is ----- type of reaction.

a)electrophilic addition

b) electrophilic substitution

c) **nucleophilic addition**

d) nucleophilic substitution

vi) The following reactants Can't be converted into carboxylic acid?

a Dry ice

**b) Cyclohexane**

c) Toluene

d) Cyclohexene

vii) The following compounds will give a positive Fehling's test.

a.Propandane

b) Pentan-3-one

c) Butanone

**d) Butan-2-ol**

viii) The tricarboxylic acid among the following is

a) oxalic acid b) malonic acid c) citric acid d) caproic acid



### Very short answer type of questions (1 mark )

1. Write IUPAC name of Phthalaldehyde?
2. Write the structure of cyclic ketal.
3. Write the name of the product when ketones react with ethane-1,2 -diol in presence of dry HCl .
4. Write another name for the disproportionation reaction?
5. Write the number of products when a mixture of ethanal and propanal is reacted with dilute alkali?
6. Write structure of the product formed when carboxylic acid is heated with dehydrating agent like  $P_2O_5$
7. Write the formula of a reducing agent, which can't reduce  $-COOH$  group of carboxylic acid.
8. Write the name of the catalyst used to convert acyl chloride into aldehyde.

### Short answer questions (Type-I) ( 2 Marks)

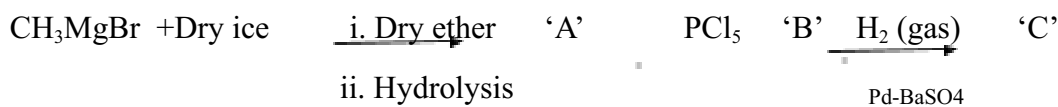
- i) Write Classification of aliphatic ketones with suitable example
- ii) What is the action of Grignard reagent on benzonitrile?
- iii) Carboxylic acids have higher boiling points than those of ethers .Give reason.
- iv) Explain Cannizzaro reaction with suitable examples.
- v) What is the action of following reagents on propanal?      a.Sodium Bisulphite    b.Hydrazine
- vi) Write the chemical reactions for the preparation of acid amide from the following.  
a.Carboxylic acid                                      b. Acid chloride
- vii) What is the action of following reagents on propanone?  
a) semicarbazide                                      b)  $Ba(OH)_2$
- viii) Write chemical equation for following conversion  
a) ethanoic acid to ethanol  
b) sodium ethanoate to methane

### Short answer questions (Type-II ) (3 marks )

- i) What is the action of following reagents on ethanoic acid?  
a)  $\text{SOCl}_2$  / heat      b) soda lime / heat      c)  $\text{P}_2\text{O}_5$  / heat
- ii) Explain aldol condensation reaction of ethanal in detail.
- iii) Write reactions for the following conversions.  
a. Benzene to Benzaldehyde  
b. Propanone to Propane  
c. 4-Nitrobenzoic acid to Nitrobenzene
- iv) Explain haloform reaction with suitable example.  
Write chemical reaction to distinguish acetaldehyde from acetone
- v) Write IUPAC name of mesityl oxide. What is the action of following reagents on acetaldehyde?  
a. hydroxylamine      b. acidified potassium dichromate
- vi) Write chemical reactions to convert  $-\text{COOH}$  group of acetic acid into following  
a.  $\text{CH}_4$    b.  $\text{C}_2\text{H}_5\text{OH}$    c.  $\text{CH}_3\text{COCl}$
- vii) Explain cross aldol condensation reaction between propanal and ethanal

### Long answer questions. (4 marks)

- i) Draw structure of salicylaldehyde.  
Write reaction for preparation of acetophenone from benzoyl chloride.  
Explain the acidic nature of carboxylic acids.
- ii) Write reactions for the action of following reagents on p-chlorobenzaldehyde.  
a) Ethane-1,2-diol in presence of dry  $\text{HCl}$ .  
b) Tollen's reagent.  
c) Phenyl hydrazine.  
Write reagent for conversion of alkyl nitrile into aldehyde.
- iii) What are aliphatic aldehydes? Complete the following sequence of reactions and write structures for A, B, C.



**Chapter -13**  
**Amines** **Marks 3 with option 4**  
**Multiple Choice Questions (1 Mark )**

- i) The following amine is the product of Gabriel phthalimide synthesis  
 a) secondary aliphatic amine                      **b) primary aliphatic amine**  
 c) primary aromatic amine                      d) tertiary aliphatic amine
- ii) Mendius' reaction is used to convert .....  
 a) amide into amine                      b) alkyl halide into amine  
 c) nitroalkane into amine                      **d) alkyl cyanide into amine**
- iii) The strongest base amongst the following is  
 a) Methanamine                      **b) N-Methylmethanamine**  
 c) N-Methylaniline                      d) N, N-Dimethylmethanamine
- iv) The reaction in which diazonium salt is used  
**a) Sandmeyer reaction**                      b) Mendius reaction  
 c) Hofmann rearrangement reaction                      d) Carbylamine reaction
- v) The type of isopropylamine is a .....  
**a) primary amine**                      b) secondary amine  
 c) tertiary amine                      d) quaternary ammonium salt
- vi) Aniline on reaction with bromine water produces .....  
 a) 1,4-Dibromobenzene                      b) 1,2,4-Tribromobenzene  
 c) 2,4-Dibromoaniline                      **d) 2,4,6-Tribromoaniline**
- vii) Tertiary amines have lowest boiling points because .....  
 a) they possess polar N-C bonds  
 b) they possess intermolecular dipole-dipole attraction forces  
**c) they possess intermolecular H-bonding**  
 d) they do not possess intermolecular H-bonding

**Very short answer type of questions (1 mark )**

1. Write the name of process of breaking the C-X bond of R-X by ammonia in preparation of amines.
2. Arrange the following compounds in increasing order of their boiling points.

Ethyl alcohol, Ethyl amine, Ethanoic acid, Ethane

3. Write the number of moles of ethanoyl chloride required for complete acylation of N,N-Dimethylaniline.
4. Write the name of the gas evolved when a primary amine is reacted with nitrous acid.
5. Write the structure of Hinsberg's reagent.
6. Write the order of reactivity of alkyl halides with ammonia.
7. Write the structure of Methylphenylamine.
8. Write the name of product formed when benzene diazonium chloride is treated with hypophosphoric acid and water.

### Short answer questions (Type-I) (2 Marks)

- i. What is the action of fluoroboric acid on arene diazonium salt and the product is heated.
- ii. Write the reactions to bring about the following conversions.
  - a. Aniline to Sulfanilic acid
  - b. Methyl chloride to Ethanamine
- iii. Explain carbylamine reaction.
- iv. Write IUPAC names of p-toluidine and Trimethylamine
- v. What is the action of nitrous acid on the following compounds?
  - a. Isopropylamine
  - b. Aniline
- vi. Write the reactions to bring about the following conversions.
  - a. Nitroalkane to Alkylamine
  - b. Alkyl cyanide to Alkylamine
- vii) What is the action of following reagents on benzenediazonium chloride ?
  - a) Phenol
  - b) aniline

### Short answer questions (Type-II) (3 marks)

- i. Distinguish between methanamine, dimethanamine and triethanamine using Hinsberg's reagent.
- ii. Write IUPAC name of  $\text{H}_2\text{N}-(\text{CH}_2)_6-\text{NH}_2$ .  
Write reactions to bring about the following conversions.
  - a) Acetamide to Ethylamine
  - b) Benzene diazonium chloride to phenol.
- iii. Explain Hoffmann's exhaustive alkylation with suitable reactions.

- iv. Explain nitration of aniline. Write reactions for the preparation of p-nitroaniline?
- v. Explain Hofmann degradation reaction with an example
- vi. Convert aniline into p-nitroaniline (indicate by reaction)

**Long answer questions (4 marks)**

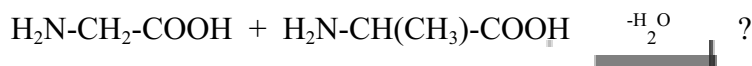
- i. Write reactions for preparation of ethanamine using Gabriel phthalimide synthesis. Why aniline cannot be prepared by Gabriel phthalimide synthesis?
- ii. Explain the basicity of amine with suitable examples. Write the factors influencing basicity of amines.
- iii. Explain the following reactions with suitable examples.
  - a) Gatterman reaction
  - b) Sandmeyer reaction.
  - c) Hofmann elimination reaction.
  - d) Hoffmann Carbylamine test.
- iv. Write the chemical reaction of ethylamine with
  - i) moist  $\text{Ag}_2\text{O}$
  - ii) Methyl iodide
  - iii) Hinsberg reagent
  - iv) Ethanoyl chloride
- v. Write the chemical reaction of Benzenediazonium chloride with following reagents
  - i) phosphinic acid(hypophosphorous acid,  $\text{H}_3\text{PO}_2$ )
  - ii) Ethanol
  - ii) Copper powder & HCl
  - iv)  $\text{CuCN/KCN}$
- vi. Do the following conversion.
  - a) Aniline into benzenediazonium chloride
  - b) benzenediazonium chloride into phenol
  - c) diethylamine into N,N-dimethylbenzenesulphonamide
  - d) methylbromide into ethylamine

**Chapter- 14**  
**Biomolecules** **Marks-3 with option 4**  
**Multiple Choice Questions (1-Mark)**

- i) Glucose on oxidation with dilute nitric acid gives-----  
a) **saccharic acid**                      b) oxalic acid                      c) gluconic acid                      d) malonic acid
- ii) The glycosidic linkage in maltose is formed between-----  
a) C-1 of  $\alpha$ -D glucose and C-2 of  $\alpha$ -D glucose    **b) C-1 of  $\alpha$ -D glucose and C-4 of  $\alpha$ -D glucose**  
c) C-1 of  $\alpha$ -D glucose and C-2 of  $\alpha$ -D fructose    d) C-1 of  $\alpha$ -D glucose and C-4 of  $\alpha$ -D fructose
- iii) The optically inactive  $\alpha$ - amino acid among the following is-----  
a) alanine                      b) insulin                      c) leucine                      **d) glycine**
- iv) The sugar component of nucleotide unit in RNA is-----  
a)  $\alpha$ - deoxy-D-ribose                      **b) D-ribose**                      c) L-ribose                      d) 2-deoxy-L-ribose
- v) The chemical nature of peptide bond in proteins is -----  
a) primary amide                      **b) secondary amide**    c) tertiary amide                      d) an ionic bond
- vi) In which of the following structure of DNA carries genetic information of the organism  
**a) the primary structure of DNA**    b) the double helix structure of DNA  
c) complementary base pairing                      d) sugar-phosphate backbone
- vii) In the process of denaturation, there is NO change in the structure following protein  
**a) primary**                      b) secondary                      c) tertiary                      d) quaternary
- viii) The base absent in DNA is  
a) adenine    b) guanine    **c) uracil**    d) thymine

**Very Short Answer Questions (1-Mark)**

- i) Write the name of polysaccharide used for commercial preparation of glucose.
- ii) Draw the structure of  $\alpha$ -D glucopyranose.
- iii) Write the structure of Zwitter ion of alanine.
- iv) Write the glycosidic linkage in sucrose.
- v) Write the name of the base present in DNA but not found in RNA.
- vi) Write the number of chiral carbon atoms present in fructose.
- vii) write the name of Sugar present in DNA.
- viii) Complete the following reaction.



ix) Draw structure of 2-deoxy-D-ribose

### Short Answer Questions (Type-I) (2-Marks)

- i) Explain preparation of glucose from sucrose
- ii) Write chemical reaction for following conversions
  - a) glucose into glucoxime
  - b) glucose into gluconic acid
- iii) Define peptide bond. Write types of proteins depending upon molecular shape.
- iv) Define the following terms.
  - a) nucleotide
  - b) nucleoside
- v) Explain denaturation of proteins
- vi) Define enzymes. Write industrial application of enzyme catalysis.
- vii) Draw structure of following.
  - a) 2-Deoxy-D-ribose
  - b) Cytosine
- viii) Explain globular and fibrous proteins with example.
- ix) Classify the following carbohydrates into monosaccharide, disaccharide, oligosaccharide and polysaccharide
 

glucose, cellulose, maltose, starch
- x) Write two chemical equations showing presence of free formyl group in glucose.
- xi) What happens when ?
  - a) glycine reacts with alanine
  - b) glucose prolonged heated with HI

### Short Answer Questions (Type-II) (3-Marks)

- i) What is the action of following reagents on glucose?
  - a) acetic anhydride
  - b) hydrogen cyanide
  - c) hydrogen iodide.
- ii) Define carbohydrates. Draw the Haworth projection structures of the following.
  - a)  $\alpha$ -D-(-) fructofuranose
  - b) maltose
- iii) Explain D and L configuration in sugars. Write a chemical reaction to convert glucose into glucose cyanohydrin.
- iv) Define  $\alpha$ -amino acids. Draw the structures of a) Zwitter ion of alanine b) Haworth formula of sucrose.
- v) Explain the primary structure of proteins. Write a commercial method for preparation of glucose.

vi) Write the structure of following

a.  $\alpha$ -D-(+) Glucose by Fischer projection formula.

b.  $\alpha$ -D-(+) Glucopyranose.

c.  $\alpha$ -D-(-) fructofuranose

vii) a) Draw the structure of enantiomers of glyceraldehyde.

b) How proteins are classified on the basis of the molecular shape.

### Long Answer Questions (4-Marks)

i) Define carbohydrates. Give the classification of carbohydrates with an example.

ii) What is monosaccharide? How is glucose prepared on a commercial scale? Draw the structure of the pyran.

iii) Write the glycosidic linkages present in Maltose, lactose, cellulose and amylose.

iv) a) Write the names of one example each of neutral and basic essential amino acids.

b) Write one example of each of non reducing and reducing sugar.



## Chapter- 15

### Introduction to polymer chemistry      Marks-3 with option 4

#### Multiple Choice Questions (1-Mark)

- i) Semisynthetic polymer among the following is -----  
a) linen                      b) silk                      c) nylon                      **d) cellulose nitrate**
- ii) Addition polymer among the following is -----  
a) terylene                      **b) polythene**                      c) nylon 6,6                      d) nylon 6.
- iii) Homopolymer among the following is -----  
a) Buna-S                      **b) Nylon 6,6**                      c) PHBV                      d) Dacron
- iv) Which of the following is a biodegradable polymer?  
a) nylon6                      b) nylon 6, 6                      **c) nylon2-nylon6**                      d) viscose rayon
- v) Chemical combination of Ziegler-Natta catalyst is-----  
a) trimethyl aluminium titanium tetrachloride  
**b) triethyl aluminium titanium tetrachloride**  
c) triethyl aluminium titanium trichloride  
d) triethyl aluminium titanium dichloride
- vi) Dacron is a copolymer of ethylene glycol and-----  
a) adipic acid                      b) hexamethylenediamine                      c) phthalic acid                      **d) terephthalic acid**
- vii) Nylon 6, 6 is a condensation polymer of hexamethylenediamine and-----  
a) picric acid                      **b) adipic acid**                      c) terephthalic acid                      d) ε caprolactam
- viii) The total number of carbon atoms are present in Nylon-6,6 polymer is /are ---  
a) 2                      b) 4                      c) 6                      **d) 12**

#### Very Short Answer Questions (1-Mark)

- i) Write the number of carbon atoms present in the monomer used for preparation of nylon 6 polymer.
- ii) Write the name of the catalyst used for preparation of high density polythene polymer.
- iii) Write the name of the monomer used for preparation of polyacrylonitrile.
- iv) Write the name of a polymer formed by condensation polymerization of monomers ethylene glycol and terephthalic acid.

- v) Write name of the monomer of natural rubber.
- vi) Write the name of biodegradable polymer formed by two amino acids namely glycine and  $\epsilon$ -amino caproic acid.
- vii) Write the name of the functional group present in terylene polymer.
- viii) Write the name of the polymer obtained by polymerization of 2-chloro-1, 3-butadiene.

### Short Answer Questions (Type-I) (2-Marks)

- i) Define elastomer. Write the name of the raw material used for preparation of nylon 6 polymer.
- ii) Write chemical reactions for the preparation of following polymers a) Teflon b) polyacrylonitrile
- iii) Define vulcanization .Write the structure of the monomer used in natural rubber.
- iv) Explain the term copolymers with examples.
- v) Write preparation of low density polythene.Mention two uses of LDP.
- vi) Write chemical reactions for preparation of the following. a) Buna-S b) Neoprene.
- vii) Explain thermoplastic and thermosetting polymers.
- viii) Explain homopolymers with examples.
- ix) Write the name of one example of each polymer in which following repeating units.  
 $(-\text{CF}_2-\text{CF}_2-)$  ,  $-\text{[NH}-(\text{CH}_2)_5\text{-CO-]}$  ,  $(-\text{CH}_2-\text{CH})-\text{CN}$ ,  $(-\text{CH}_2-\text{CH}_2-)$
- x) How is polythene prepared? Write their types.

### Short Answer Questions (Type-II) (3-Marks)

- i) Explain classification of polymers on the basis of structures.
- ii) Explain copolymers. Write the name and formulae of the monomers used for preparation of Dacron.
- iii) Write chemical reactions for the preparation of high density polythene.Write its two uses and two properties.
- iv) Write the preparation of nylon 6, 6. Mention two properties and two uses of nylon 6,6 polymer.
- v) Explain classification of polymers on the basis of origin.
- vi) Define fibres.Explain vulcanization of rubber.
- vii) Explain free radical mechanisms in detail for the preparation of addition polymers.
- viii) Write three stages preparation of bakelite
- ix) Write commercial importance of a) Buna-S b) PVC c) Urea formaldehyde resin.

### Long Answer Questions (4-Marks)

- i) Define rubber. Write the chemical reactions for the preparation of following polymers.  
a) teflon   b) polyisoprene   c) polyacrylonitrile   d) SBR
- ii) Explain the reactions involved in the preparation of viscose rayon.
- iii) What are addition polymers. Write free radical mechanism of addition polymerisation.

## Chapter-16

### Green Chemistry and Nanochemistry

Marks 3 with option 4

#### Multiple choice questions (1 Mark)

- i) Bottom ash of thermal power stations can be used as raw material for cement and brick industry. This example illustrates which of the following principle of green chemistry
- |                                  |                                               |
|----------------------------------|-----------------------------------------------|
| a) Atom economy.                 | b) Designing safer chemicals.                 |
| c) Design for energy efficiency. | <b>d) Prevention of waste or by products.</b> |
- ii) Less hazardous chemical synthesis point of view instead of harmful DDT Now a days ----- is used as insecticides
- |                  |               |
|------------------|---------------|
| a) Benzene       | <b>b) BHC</b> |
| c) Chlorobenzene | d) Ethanol    |
- iii) The concept that aims to maximize efficiency and minimize hazardous effect on human health and environment was coined by Paul T. Anastas
- |                     |                           |
|---------------------|---------------------------|
| a) Green revolution | b) Blue revolution        |
| c) Nano chemistry   | <b>d) Green Chemistry</b> |
- iv) Nanorods are the example of -----
- |                                    |                                         |
|------------------------------------|-----------------------------------------|
| a) One dimensional nanostructure   | <b>b) Two Dimensional nanostructure</b> |
| c) Three dimensional nanostructure | d) Zero dimensional nanostructure       |
- v) Which nanoparticles act as highly effective bacterial disinfectants, removing E.Coli from water?
- |                                 |                                |
|---------------------------------|--------------------------------|
| a) Gold nanoparticles           | <b>b) Silver nanoparticles</b> |
| c) $\text{TiO}_2$ nanoparticles | d) $\text{ZnO}$ nanoparticles  |
- vi) Catalyst used for hydrogenation of oil is-----
- |                           |                   |
|---------------------------|-------------------|
| a) $\text{V}_2\text{O}_5$ | b) Fe             |
| <b>c) Raney Ni</b>        | d) $\text{MnO}_2$ |
- vii) In green technology developed by Drath and Frost, adipic acid is enzymatically synthesized from
- |            |                   |
|------------|-------------------|
| a) Sucrose | b) lactose        |
| c) maltose | <b>d) glucose</b> |

viii) One nanometer is equal to –

- a)  $10^{-3}$  m      b)  $10^{-4}$  m      c)  $10^{-5}$  m      d)  $10^{-9}$  m

### Very short answer Questions (1 Mark)

- i) Write the name of the catalyst used to manufacture of  $\text{H}_2\text{SO}_4$  by contact process.
- ii) Write the name of the plant which is an example of self-cleaning.
- iii) Write the name of a metal nanomaterial, which is used as a bacterial disinfectant.
- iv) Name the  $\gamma$ -isomer of Benzene hexachloride
- v) Which principle of green chemistry has its perspective towards to carrying out reactions at room temperature and pressure
- vi) Write any example of nanoparticles which are used in photo catalysis.
- vii) Write the name of the technique used to determine the geometry of nanomaterial.
- viii) Write the name of the nanostructured material that is used to increase the life of a car tyre.
- ix) Write the name of a nanomaterial synthesis process based on inorganic Polymerization.

### Short Answer Questions (Type-I) (2-Marks)

- 1) Define: Atom economy. Write the formula to calculate % atom economy
- 2) Define      i) Green chemistry      ii) Nanochemistry
- 3) Write one example of safer solvent and hazardous solvent
- 4) How does nanochemistry play an important role in water purification?
- 5) Write advantages of nanotechnology with respect to
  - a) energy sector
  - b) medical field

### Short Answer Questions (Type-II) (3-Marks)

- 1) Explain prevention of waste or by-products which is one of the principles of green chemistry.
- 2) Write any three advantages of nanoparticles and nanotechnology.

- 3) Write one example of nanomaterial used in following
  - i) water purification
  - ii) tyre of car
  - iii) ancient glass painting
- 4) Explain the role of green chemistry.
- 5) Explain any three characteristic features of nanoparticles.
- 6) State Disadvantages of nanoparticles and nanotechnology.
- 7) Define : a) Green chemistry b) Atom economy c) Sustainable development
- 8) Write three principles of green chemistry with examples.
- 9) Mention any three principles of green chemistry.
- 10) Find out atom economy of following conversion
  - a) 1-chlorobutane to butan-1-ol
  - b) write the name of metal oxide used in sunscreen lotions (creams)

### Long Answer Questions (4-Marks)

- 1) i) Explain the term sustainable development  
 ii) How is nanotechnology useful for the energy sector?
- 2) i) Write a short note on catalytic activity of nanoparticles.  
 ii) Complete and write the following table

| S/N | Nanomaterial dimension | Nanomaterial type |
|-----|------------------------|-------------------|
| a)  | One dimension <100 nm  | -----             |
| b)  | Two dimension <100 nm  | -----             |

- 3) i) Explain use of safer solvent by giving suitable examples.  
 ii) Define a) Nanomaterial b) Nanotechnology